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$$\begin{aligned}f(x) &= \frac{x^n + 1}{x^n - 1} \text{ met } n \in \mathbb{N}_0 \\f'(x) &= \frac{(x^n - 1) \cdot nx^{n-1} - (x^n + 1) \cdot nx^{n-1}}{(x^n - 1)^2} \\&\Rightarrow f'(x) = \frac{nx^{n-1}(x^n - 1 - x^n - 1)}{(x^n - 1)^2} \\&\Rightarrow f'(x) = \frac{nx^{n-1} \cdot (-2)}{(x^n - 1)^2} \\&\Rightarrow f'(x) = \frac{-2nx^{n-1}}{(x^n - 1)^2}\end{aligned}$$

References